

PhD Defense:

Random Matrix Theory for Linearized Neural Networks



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# Acknowledgments

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- Chapter 3 is extracted from “Collin Cranston, Zhichao Wang, Todd Kemp, Michael W. Mahoney. Eigen-Spike Emergence and Quadratic Equivalents for Conjugate Kernels on Nonlinearly Separable Data, Proceedings of the 39th Annual Conference on Learning Theory, *Proceedings of Machine Learning Research (PMLR)*. arXiv:2605.29669 “ To appear.



Todd Kemp



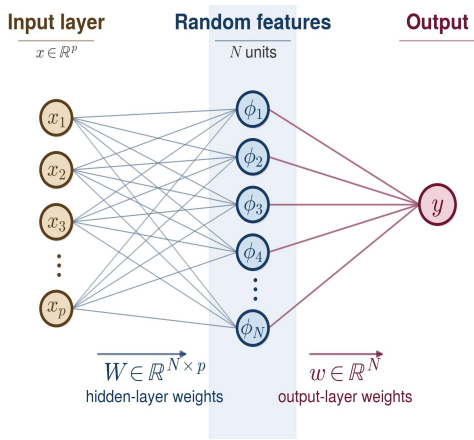
Zhichao Wang



Michael Mahoney

# Abstract

- Nonlinear Random Matrix Theory (RMT) and Free Probability in Deep Learning.
- Universality result, called *Gaussian Equivalence*, allows for the behavior of complex nonlinear neural networks to be understood through tractable linear kernel models.
- This motivates the question whether this idealized linear equivalence remains meaningful beyond isotropic Gaussian setting.



Single hidden-layer Neural Network

## Chapter 2: NTK Regression under dual power-law model: Deterministic Equivalents via SDE and PDE Methods

- Model settings and motivation

- Outline of approach

- Main results

- Experiments

## Chapter 3: Eigen-Spike Emergence and Quadratic Equivalents for Conjugate Kernels on Nonlinearly Separable Data

- Background and motivation

- Main results

- Experiments

Conclusion

**Chapter 2: NTK Regression under  
dual power-law model:  
Deterministic Equivalents via SDE  
and PDE Methods**

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# Two-Layer Neural Network

Consider a **two-layer Neural Network**

$$f_{\theta}(\mathbf{x}) = \frac{1}{\sqrt{N}} \mathbf{w}^{\top} \sigma(\mathbf{W}_1 \mathbf{x}), \quad \theta := (\mathbf{w}_1^{(1)}, \dots, \mathbf{w}_N^{(1)}, w_1, \dots, w_N) \in \mathbb{R}^{N(p+1)}$$

- $\mathbf{x} \in \mathbb{R}^p$  is the input data,  $p$  is the ambient feature dimension,  $N$  is the network width.
- $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is the entry-wise nonlinear activation function (e.g ReLU, tanh, sigmoid).
- First layer weights  $\mathbf{W}_1 \in \mathbb{R}^{N \times p}$  are initialized as  $(\mathbf{W}_1)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, p^{-1})$ .
- Second layer weights  $\mathbf{w} \in \mathbb{R}^N$  are initialized as  $w_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_W^2)$

## Lazy Training Regime

When the width  $N \rightarrow \infty$ , the weights  $\theta$  stay close to their initialization  $\theta_0$  throughout training via gradient descent. Moreover, the model output  $f_{\theta}(\mathbf{x})$  is well-approximated by a first-order Taylor expansion around  $\theta_0$ . [JGH18]

**Linearize** the neural network, i.e perform a first-order Taylor expansion around  $\theta_0$ .

$$f_{\theta}(\mathbf{x}) \approx f(\mathbf{x}; \theta_0) + (\theta - \theta_0)^{\top} \nabla_{\theta} f(\mathbf{x}; \theta_0)$$

- $f(\mathbf{x}; \theta_0)$  denotes the network output at initialization.
- Perform centering by setting  $f(\mathbf{x}; \theta_0) = 0$  [AP20].

## Neural Tangent Kernel (NTK)

The (finite-width) NTK is the gram matrix of the Jacobian of the network output respect to weights  $\theta$ :

$$\mathbf{K}^{\text{NTK}}(\mathbf{x}, \mathbf{x}') := \nabla_{\theta} f(\mathbf{x}; \theta_0)^{\top} \nabla_{\theta} f(\mathbf{x}'; \theta_0)$$

For data matrix  $\mathbf{X} \in \mathbb{R}^{p \times n}$ ,

$$\mathbf{K}^{\text{NTK}} = \mathbf{J}^{\top} \mathbf{J}, \quad \mathbf{J} := \nabla_{\theta} f(\mathbf{X}; \theta_0) \in \mathbb{R}^{N(p+1) \times n}$$

# Why study the NTK?

Introduced to study training dynamics of ultra-wide Neural Networks (NNs) through tractable kernel models [JGH18].

- Under gradient-flow dynamics, the NTK is approximately constant and training reduces to linear regression in tangent features.
- Spectral properties of NTK govern generalization performance of NNs.
- We focus on the finite-width realization of NTK for its connection to NNs of practical width.
- Scaling laws<sup>1</sup> in finite-width NTK remain largely open theoretically.

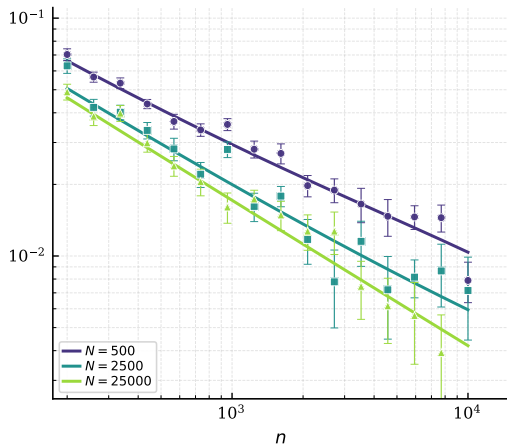
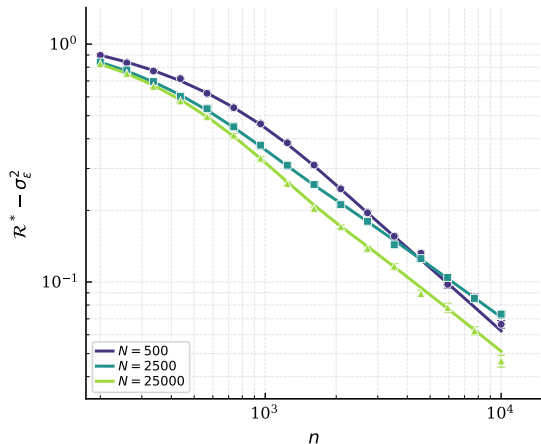
$$\mathcal{L}(n, N) = \left[ \left( \frac{N_c}{N} \right)^{\alpha_N / \alpha_n} + \left( \frac{n_c}{n} \right) \right]^{\alpha_n}$$

**Key references:** Jacot et al. (2018), Chizat et al. (2019), Kaplan et al. (2020), Anwar et al. (2024)

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<sup>1</sup>Model performance scales as a power-law in samples  $n$  and parameters  $N$ . (Kaplan et al. 2020)

# Scaling Law



Optimally-tuned excess NTK risk: Plot versus sample size  $n$  with log-log axis. **(Left)** slowly-decaying spectrum  $\alpha = 0.2$ , with visible curvature on log-log plot. **(Right)** rapidly-decaying spectrum  $\alpha = 0.8$ , exhibiting a power-law decay in  $n$ . Empirical test loss reported as markers  $\pm$ SE over 20 trials. Plots generated with  $r = 1.5, p = 1200, \sigma_\epsilon^2 = 0.3, \sigma_W^2 = 1.0$

## Learning Problem:

- Goal: learn unknown target function  $f^* : \mathbb{R} \rightarrow \mathbb{R}$  from  $n$  training samples  $\{(x_i, y_i)\}_{i=1}^n \in \mathbb{R}^p \times \mathbb{R}$ .
- Student-teacher model:  
 $y_i = f^*(x_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$
- Consider model defined by the linearization of  $f_\theta(\cdot)$  around  $\theta_0$ .
- Perform (approximate) kernel ridge regression (KRR) via the finite-width NTK with ridge parameter  $\lambda > 0$ .

## NTK Predictor:

$$\hat{f}_\theta^\lambda(\mathbf{x}) = \mathbf{k}_{\text{NTK}}(\mathbf{x})^\top (\mathbf{K}^{\text{NTK}} + \lambda \mathbf{I}_n)^{-1} \mathbf{y}$$

- $\mathbf{k}_{\text{NTK}}(\mathbf{x}) \in \mathbb{R}^n$  has entries  $\mathbf{K}^{\text{NTK}}(\mathbf{x}, \mathbf{x}_\alpha)$  for  $\alpha \in [n]$ .
- Ridge parameter  $\lambda > 0$  corresponds to  $\ell^2$ -regularization of parameter weights towards initialization.
- As  $N \rightarrow \infty$ , converges to deterministic kernel that describes network trajectory during training under gradient descent.

## Training:

- **Empirical Risk minimization (ERM)** of  $\ell^2$ -regularized Mean Squared Error (MSE):

$$L_\lambda(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \left( y_i - \hat{f}_\theta^\lambda(\mathbf{x}_i) \right)^2 + \frac{\lambda}{n} \|\boldsymbol{\theta}\|_2^2$$

- **Closed-form solution:** Optimal kernel ridge predictor:

$$\hat{\boldsymbol{\beta}}_\lambda = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} L_\lambda(\boldsymbol{\beta}) = \mathbf{k}_{\text{NTK}}(\mathbf{x})^\top (\mathbf{K}^{\text{NTK}} + \lambda \mathbf{I}_n)^{-1} \mathbf{y}$$

- **Training error** under optimal predictor  $\hat{\boldsymbol{\beta}}_\lambda$ :

$$\mathbb{E}_{\text{train}} = \frac{1}{n} \|\mathbf{y} - \hat{f}_\theta^\lambda(\mathbf{X})\|^2 = \frac{\lambda^2}{n} \mathbf{y}^\top (\mathbf{K}^{\text{NTK}} + \lambda \mathbf{I}_n)^{-2} \mathbf{y}$$

## Testing:

- Prediction risk:

$$\mathcal{R}(\lambda) = \mathbb{E}_{\mathbf{x}}[(\hat{f}_{\theta}^{\lambda}(\mathbf{x}) - y)^2]$$

- Expectation is over an independent test point  $(\mathbf{x}, y) \in \mathbb{R}^p \times \mathbb{R}$  conditional on training dataset and random weights.
- Random variable, but will concentrate around its mean (over  $\mathbf{X}, \varepsilon$ ) in the proportional asymptotic regime.

## Main goal

Derive deterministic equivalents<sup>2</sup> for the random variables

$$\mathbb{E}_{\text{train}} = \frac{1}{n} \|\mathbf{y} - \hat{f}_{\theta}^{\lambda}(\mathbf{X})\|_2^2, \quad \mathcal{R}(\lambda) = \mathbb{E}_{\mathbf{x}}[(y - \hat{f}_{\theta}^{\lambda}(\mathbf{x}))^2]$$

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<sup>2</sup>finite-dimensional deterministic surrogates that approximate the random matrix under scalar functionals (traces, bilinear forms)

# Dual-Power Law Model

Focus on dataset that allows for **scaling laws** in the excess prediction risk  $\mathcal{R}(\lambda) - \sigma_\varepsilon^2$ .

## (A0): Dual Power-Law Model

The training data  $\mathbf{x}_1, \dots, \mathbf{x}_n$  are drawn i.i.d from  $\mathcal{N}(0, \Sigma)$  where

$$\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p), \quad \sigma_j = j^{-\alpha}, \quad j \in [p] \quad (1.1)$$

for some fixed  $\alpha \in (0, \infty)$ . The labels are generated as a linear power-teacher.

$$\mathbf{y}_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \beta_j = j^{-r}, \quad j \in [p] \quad (1.2)$$

where  $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$  and  $r > 0$  is a fixed constant.

- **Capacity** exponent  $\alpha$  controls effective dimensionality of data, with larger values causing lower effective dimensionality.
- **Source** exponent  $r > 0$  controls difficulty of learning problem, measures alignment of signal  $\boldsymbol{\beta}$  onto top eigenvectors of  $\Sigma$ .

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where  $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$  and  $r > 0$  is a fixed constant.

- *Signal captured by coordinate  $j$  is given by  $\beta_j^2 \sigma_j \sim j^{-(2r+\alpha)}$ ; the exponent  $2r + \alpha$  dictates the scaling rate.*
- Entries of  $\mathbf{W}_1 \mathbf{X}$  are marginally  $\mathcal{N}(0, \tau^2)$ , where  $\tau^2 := \frac{1}{p} \text{Tr } \Sigma$ .

# Main Assumptions

We make the following general assumptions.

## Assumptions

- (A1) Proportional Regime.**  $n, p, N \rightarrow \infty$  at rates  $\frac{n}{p} = \phi$  and  $\frac{N}{p} = \psi$  for some  $\phi, \psi \in (0, \infty)$ .
  - (A2) Capacity Exponent.** We take  $\Sigma_{jj} = j^{-\alpha}$  with  $\alpha > 0$ .
  - (A3) Source Exponent.** The signal vector satisfies  $\beta_j = j^{-r}$  with  $r > \frac{1}{2}$ .
  - (A4) Nonlinearity.** The non-linearity  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is Lipschitz, centered with respect to  $\mathcal{N}(0, \tau^2)$ , and has finite Gaussian moments of all order.
- Assumption (A3) ensures  $\|\beta\|_2 < \infty$  as  $p \rightarrow \infty$ .
  - Assumption (A4) allows for most common activation functions (after centering), e.g ReLU, sigmoid, tanh, softplus.

We apply three **simplifying principles**

## Simplifying Principles

- (C1) **Gaussian Equivalence I**: Replace nonlinear matrices with **linear** Gaussian surrogates with matching first and second order moments.
- (C2) **Gaussian Equivalence II**: Asymptotic training and test error of NTK regression are asymptotically equivalent to that of regression on **linearized** kernel.
- (C3) **Generalized Cross Validation (GCV)**: GCV is a uniformly consistent estimator of prediction risk.

We **conjecture** (C1)-(C3) extend to the anisotropic power-law setting.

**Key references**: Adlam et al. (2019), Hu–Lu (2022), Montanari–Saeed (2022), Peche et al. (2019)

**Universality principle** from RMT in proportional asymptotic ( $n \asymp N \asymp p$ ) analysis of nonlinear kernel models.

## Gaussian Equivalence I

$$\sigma(\mathbf{W}_1 \mathbf{X}) \approx b_1 \mathbf{W}_1 \mathbf{X} + b_2 \Theta, \quad \Theta_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

when  $\mathbb{E}[\sigma(g)] = 0$ , where  $b_1 = \mathbb{E}[\sigma'(g)]$ ,  $b_2^2 = \mathbb{E}[\sigma(g)^2] - \tau^2 b_1^2$  and  $g \sim \mathcal{N}(0, \tau^2)$ .

## Gaussian Equivalence II

The following holds pointwise for each fixed  $\lambda \geq 0$ .

$$|\mathcal{R}(\lambda) - \mathcal{R}_{\text{GE}}(\lambda)| = o_{n, \mathbb{P}}(1)$$

After further simplification, we define the **linearized** NTK. Let  $c_1 = \mathbb{E}[\sigma'(g)^2]$

$$\mathbf{K}_{\text{lin}} = \mathbf{F}_{\text{lin}}^\top \mathbf{F}_{\text{lin}} + \sigma_W^2 \left( p^{-1} b_1^2 \mathbf{X}^\top \mathbf{X} + \tau^2 (c_1 - b_1^2) \mathbf{I}_n \right) + z \mathbf{I}_n, \quad \mathbf{F}_{\text{lin}} = \frac{1}{\sqrt{N}} (b_1 \mathbf{W}_1 \mathbf{X} + b_2 \Theta)_{16/81}$$

**Generalized Cross Validation (GCV):** Introduced as rotationally-invariant approximation of LOOCV [GHW79].

## GCV

$$\text{GCV}(\lambda) := \frac{E_{\text{train}}(\lambda)}{\left(\lambda \text{tr}(\mathbf{K}^{\text{NTK}} + \lambda \mathbf{I}_n)\right)^2}$$

- In-sample estimator: agnostic to true data distribution.
- Consistent estimator of prediction risk for wide class of kernel models and their associated learning problems.
- Shown to approximate prediction risk of pretrained Residual Neural Network (RNN) on dataset of images (CIFAR-10) [WHS22].

### Uniform Consistency of GCV

$$\sup_{\lambda \in [\lambda_{\min}, \lambda_{\max}]} |\mathcal{R}(\lambda) - \text{GCV}(\lambda)| = o_{n, \mathbb{P}}(1)$$

- Proved for KRR under general concentration assumptions on features, which covers our dual power-law model, where a rate of  $\mathcal{O}_{n, \mathbb{P}}\left(n^{-\frac{1}{2}}\right)$  was achieved [Mis22].
- Extends to  $\lambda_{\min} = 0$  in benign overfitting regime, which allows for optimal ridge-tuning via GCV.
- **Conjecture** GCV is a uniformly consistent estimator for prediction risk of finite-width NTK regression on power-law data.

**Key references:** Adlam–Pennington (2020), Misiakiewicz–Saeed (2024), Patil et al. (2021)

### Asymptotic Training Error

$$E_{\text{train}}(\lambda) = \lambda^2 \left( \sigma_\varepsilon^2 \text{tr}(\mathbf{K} + \lambda \mathbf{I}_n)^{-2} + \text{tr}(\mathbf{X} \boldsymbol{\beta} \boldsymbol{\beta}^\top \mathbf{X}^\top (\mathbf{K} + \lambda \mathbf{I}_n)^{-2}) \right) + o_{n, \mathbb{P}}(1)$$

Define

$$m_1^{(n)}(\lambda) = \text{tr}(\mathbf{K} + \lambda \mathbf{I}_n)^{-1}, \quad m_2^{(n)}(\lambda) = \text{tr}(p^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{K} + \lambda \mathbf{I}_n)^{-1})$$

$$m_\beta^{(n)}(\lambda) = \frac{1}{n} (\boldsymbol{\beta}^\top \mathbf{X}^\top (\mathbf{K} + \lambda \mathbf{I}_n)^{-1} \mathbf{X} \boldsymbol{\beta})$$

### Asymptotic Training Error corr.

$$E_{\text{train}}(\lambda) = -\lambda^2 \left( \sigma_\varepsilon^2 \partial_\lambda m_1^{(n)}(\lambda) + \partial_\lambda m_\beta^{(n)}(\lambda) \right) + o_{n, \mathbb{P}}(1)$$

### GCV corr.

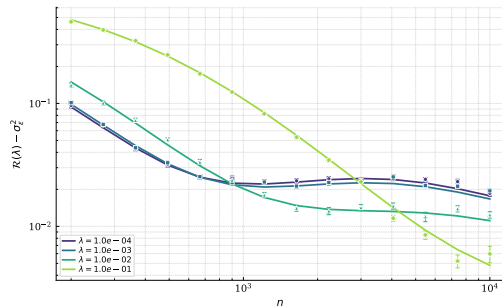
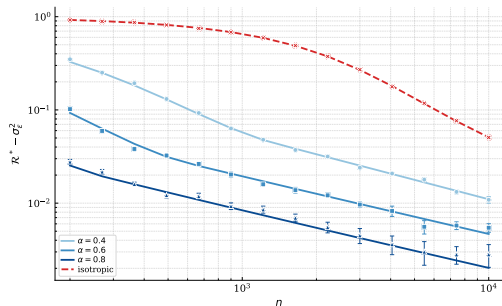
$$\text{GCV}(\lambda) = \frac{-(\sigma_\varepsilon^2 \partial_\lambda m_1^{(n)}(\lambda) + \partial_\lambda m_\beta^{(n)}(\lambda))}{(m_1^{(n)}(\lambda))^2}$$

- GCV can be understood as correction to train error to account for model complexity.
- $m_1^{(n)}(\lambda)$  accounts for train-test gap, measures effective degrees of freedom of linear estimator onto  $\mathbf{K}^{\text{NTK}}$ .
- **Signal resolvent**  $m_\beta^{(n)}$  measures alignment of signal  $\beta$  with resolvent of  $\mathbf{K}^{\text{NTK}}$ .
- $m_\beta^{(n)}(\lambda) \approx m_2^{(n)}(\lambda)$  concentrates under isotropic random signal  $\beta \sim \mathcal{N}(0, p^{-1} \mathbf{I}_p)$ .

### New Goal

Derive deterministic equivalents to  $m_1^{(n)}(\lambda)$ ,  $m_2^{(n)}(\lambda)$  and  $m_\beta^{(n)}(\lambda)$

# Isotropic vs power law



**Isotropic vs power-law:** **(Left)** Optimally-tuned excess NTK risk plotted against  $n$  for varying  $\alpha$  with isotropic baseline in red. Isotropic case exhibits curve in log-log axes plot, while power law curve straightens as  $\alpha$  increases. **(Right)** excess risk against  $n$  for various ridge  $\lambda$  with fixed  $\alpha = 0.6$ . Both plots generated with  $p = 2000, \sigma_\epsilon^2 = 0.3, r = 1.5, N = 2500, \sigma_W^2 = 1.0$ . Error bars are plotted with marker  $\pm$  SE over 20 trials.

**Main Technical Tool:** Log-determinant of **Brownian Motion** embedded NTK matrix.

## Log-Determinant Potential

$$\mathcal{F}_n(t, \kappa, \gamma) = \frac{1}{n} \log \det \mathbf{K}(t, \kappa, \gamma)$$

Recall **linearized** NTK matrix

$$\mathbf{K} = \mathbf{F}^\top \mathbf{F} + \sigma_W^2 (p^{-1} b_1^2 \mathbf{X}^\top \mathbf{X} + \tau^2 (c_1 - b_1^2) \mathbf{I}_n) + z \mathbf{I}_n, \quad \mathbf{F} = \frac{1}{\sqrt{N}} (b_1 \mathbf{W}_1 \mathbf{X} + b_2 \mathbf{\Theta})$$

**Approach idea:** Condition on train data  $\mathbf{X}$ , embed each Gaussian-entry matrix with i.i.d Brownian motions

$$\mathbf{W}_1 \leftrightarrow \mathbf{W}_1(\mathbf{t}), \quad \mathbf{\Theta} \leftrightarrow \mathbf{\Theta}(\mathbf{t}), \quad \mathbf{F} \leftrightarrow \mathbf{F}(\mathbf{t})$$

where

$$\mathbf{A}(\mathbf{t}) \Leftrightarrow \mathbf{A}_{ij}(\mathbf{t}) \sim \text{BM}(0, 1)$$

## Define

### Time-parametrized NTK

$$\mathbf{K}(t, \kappa, \gamma) = \mathbf{F}(t)^\top \mathbf{F}(t) + p^{-1} \kappa \mathbf{X}^\top \mathbf{X} + \gamma \mathbf{I}_n, \quad \mathbf{F}(t) = \frac{1}{\sqrt{N}} (b_1 \mathbf{W}_1(t) \mathbf{X} + b_2 \boldsymbol{\Theta}(t))$$

- $\kappa = \kappa(t)$  and  $\gamma = \gamma(t)$  are parameters, with target values

$$\kappa_{\text{target}} := \sigma_W^2 b_1^2, \quad \gamma_{\text{target}} = \lambda + \sigma_W^2 \tau^2 (c_1 - b_1^2)$$

- **Motivation:**

$$\frac{\partial \mathcal{F}_n}{\partial \gamma} = \underbrace{\text{tr}(\mathbf{K}(t, \kappa, \gamma)^{-1})}_{m_1^{(n)}(t, \kappa, \gamma)}, \quad \frac{\partial \mathcal{F}_n}{\partial \kappa} = \underbrace{\text{tr}\left(p^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{K}(t, \kappa, \gamma)^{-1})\right)}_{m_2^{(n)}(t, \kappa, \gamma)}$$

**Step 1:** Apply entry-wise Itô-calculus to  $\mathcal{F}_n$  to derive a finite-dimensional Stochastic Differential Equation (SDE) of the form<sup>3</sup>

$$d\mathcal{F}_n(t, \mathbf{u}) = d\mathcal{M}_n(t, \mathbf{u}) + \mathcal{D}_n(t, \mathbf{u})dt + E_n(t)$$

**Step 2:** Control the Martingale terms  $\mathcal{M}_n(t, \mathbf{u})$  and remainder terms  $E_n(t)$  uniformly in  $t \in [0, 1]$ , recognize drift terms  $\mathcal{D}_n(t, \mathbf{u})$  as partial derivatives to derive an asymptotic deterministic partial differential equation (PDE) of the form

$$\partial_t \mathcal{F}_\infty(t, \mathbf{u}) + H(\mathbf{u}, \partial_{\mathbf{u}} \mathcal{F}_\infty) = 0 \tag{1.3}$$

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<sup>3</sup> $\mathbf{u} := (\kappa, \gamma)$

**Step 3:** Recognize PDE (1.3) as a first-order nonlinear PDE of **Hamilton-Jacobi (HJ)** type.

$$\partial_t \mathcal{F}_\infty(t, \mathbf{u}) + H(\mathbf{u}, \mathbf{p}_\mathbf{u}) = 0$$

where the Hamiltonian  $H$  is obtained from replacing partial derivatives  $\partial_{u_j}$  with corresponding momentum variables  $p_j$ .

**Step 4:** Solve HJ PDE using Method of Characteristics. Hamilton's equations translate PDE into system of ordinary differential equations (ODEs)

$$\dot{\mathbf{u}}_j = \partial_{p_j} H, \quad \dot{p}_j = -\partial_{\mathbf{u}_j} H$$

with initial conditions defined by  $\mathcal{F}_\infty(0, \mathbf{u}(0))$ . Momentum variables  $p_j$  will correspond to partial derivatives of  $\mathcal{F}_\infty(t)$  along curves  $\mathbf{u}_j$ .

## Proposition: Asymptotic PDE

Under Assumptions (A1) and (A4),  $\mathcal{F}_n(t, \kappa, \gamma)$  *approximately* satisfies the following PDE

$$\partial_t \mathcal{F} = (b_1^2 \partial_\kappa \mathcal{F} + b_2^2 \partial_\gamma \mathcal{F}) \left[ \left(1 - \frac{\phi}{\psi}\right) + \frac{\phi}{\psi} (\kappa \partial_\kappa \mathcal{F} + \gamma \partial_\gamma \mathcal{F}) \right] \quad (1.4)$$

### Next Steps:

- Derive unique analytic solution to (1.4) with initial condition

$$\mathcal{F}_\infty(0, \kappa_0, \gamma_0) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \det(p^{-1} \kappa_0 \mathbf{X}^\top \mathbf{X} + \gamma_0 \mathbf{I}_n)$$

- $\kappa_0, \gamma_0$  are initial values of  $(\kappa(t), \gamma(t))$  characteristic curves, chosen to reach target values  $(\kappa_{\text{target}}, \gamma_{\text{target}})$  at time  $t = 1$ .
- Deterministic equivalents for initial momentum variables  $p_\kappa(0), p_\gamma(0)$  are given by the general **Marchenko-Pastur Theorem** [BS10].

## Main Theorem Setup:

Denote the following,  $D_\lambda = \lambda + \tau^2 \sigma_W^2 (c_1 - b_1^2)$ . Then for  $\eta > 0$ , define the rational functions

$$s(\eta) = \frac{b_1^2(\eta + \sigma_W^2)}{\eta(D_\lambda - b_2^2 \sigma_W^2)} (\phi\eta - \psi(1 - \eta)(\eta + \sigma_W^2)), \quad \zeta(\eta) = \frac{b_2^2 \eta + D_\lambda}{b_1^2(\eta + \sigma_W^2)}$$

Let  $\eta^* > 0$  be the unique positive solution to the fixed point equation

$$\Psi(\eta^*) = F_\alpha(s(\eta^*)) + \zeta(\eta^*)s(\eta^*) - \phi = 0, \quad F_\alpha(s) = \frac{1}{p} \sum_{j=1}^p \frac{s}{s + j^\alpha} \quad (1.5)$$

- The fixed point equation (1.5) has a unique physical solution  $\eta^* > 0$  corresponding to  $s(\eta^*) > 0$ , and can be found via standard numerical methods (e.g bracketing via brentq [Bre73]).

## Theorem: Power-Law Resolvents

Under Assumptions (A0)-(A4), we have

$$m_1(\lambda) \asymp \frac{1}{\phi b_1^2(\eta^* + \sigma_W^2)} s(\eta^*), \quad m_2(\lambda) \asymp \frac{1}{\phi b_1^2(\eta^* + \sigma_W^2)} F_\alpha(s(\eta^*))$$

where  $F_\alpha(s) = \frac{1}{p} \sum_{j=1}^p \frac{s}{s+j^\alpha}$ . Moreover,

$$m_\beta(\lambda) \asymp \frac{1}{\phi b_1^2(\eta^* + \sigma_W^2)} F_r(s(\eta^*)), \quad F_r(s) = \sum_{j=1}^p \frac{s j^{-(\alpha+2r)}}{s j^{-\alpha} + 1}$$

## Remarks

- Resolvent traces are characterized as momentum curves of HJ PDE  $p_\kappa, p_\gamma, p_\xi$ . The full rigorous justification of the identifications  $m_1^\infty = m_1, m_2^\infty = m_2$ , and  $m_\beta^\infty = m_\beta$  is beyond scope of this work.

## Remarks

- Signal resolvent  $m_\beta$  is the momentum curve  $p_\xi$  of **augmented** HJ PDE:

$$\tilde{\mathcal{F}}_n(t, \lambda, \kappa, \xi) = n^{-1} \log \det \left( \mathbf{K}(t, \kappa, \gamma) + \xi \mathbf{X}^\top \beta \beta^\top \mathbf{X} \right)$$

This is a **rank-1** perturbation of original log-determinant potential, recover original HJ along  $\xi = 0$  slice.

- Resolvent traces take the form

$$m_1 = \frac{p_\gamma(0)}{\eta^*}, \quad m_2 = \frac{p_\kappa(0)}{\eta^*}, \quad m_\beta = \frac{p_\xi(0)}{\eta^*}$$

Reduces coupled polynomial system of [AP20] and fixed-point equations of [MM19] to single scalar-root solver.

## Remarks

- Signal resolvent  $m_\beta$  is the momentum curve  $p_\xi$  of **augmented** HJ PDE:

$$\tilde{\mathcal{F}}_n(t, \kappa, \gamma) = \mathcal{F}_n(t, \kappa, \gamma) + \frac{1}{n} \log(1 + \xi \mathbf{u}^\top \mathbf{K}(t, \kappa, \gamma)^{-1} \mathbf{u})$$

This is a **rank-1** perturbation of original log-determinant potential, recover original HJ along  $\xi = 0$  slice.

- Resolvent traces take the form

$$m_1 = \frac{p_\gamma(0)}{\eta^*}, \quad m_2 = \frac{p_\kappa(0)}{\eta^*}, \quad m_\beta = \frac{p_\xi(0)}{\eta^*}$$

Reduces coupled polynomial system of [AP20] and fixed-point equations of [MM19] to single scalar-root solver.

## Proposition: Prediction Risk

Under Assumptions (A0)-(A4), the prediction risk admits a bias-variance decomposition

$$\mathcal{R}(\lambda) \asymp \mathcal{B}(\lambda; \alpha, r) + \mathcal{V}(\lambda; \alpha, r) \quad (1.6)$$

where the bias and variance terms are defined as

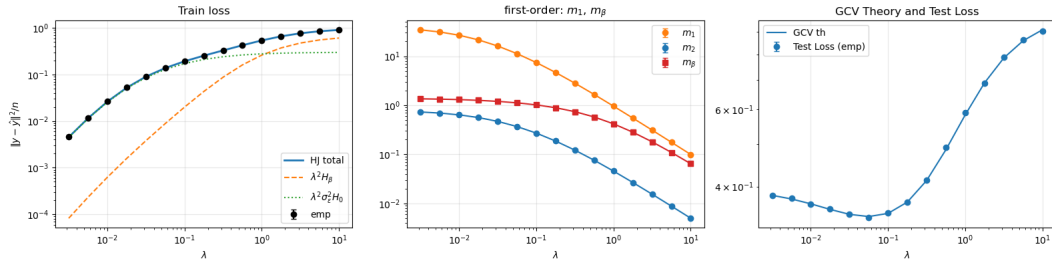
$$\mathcal{B}(\lambda; \alpha, r) = -\frac{\partial_\lambda m_\beta(\lambda)}{m_1(\lambda)^2}, \quad \mathcal{V}(\lambda; \alpha, r) = -\frac{\sigma_\varepsilon^2 \partial_\lambda m_1(\lambda)}{m_1(\lambda)^2}$$

### Remarks:

- Follows from implicit differentiation of resolvent traces from main theorem and the GCV estimator, full formulas omitted.
- Bias term depends on **source** strength  $r$  through  $m_\beta$ , while variance term depends *only* on **capacity**  $\alpha$  through  $m_1$ .

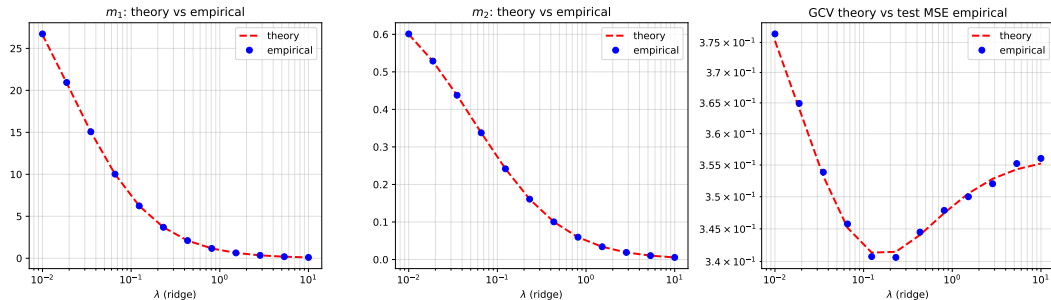
# Experiments

$m_1$ ,  $m_2$ , and  $m_\beta$  with power covariance, and power teacher labels



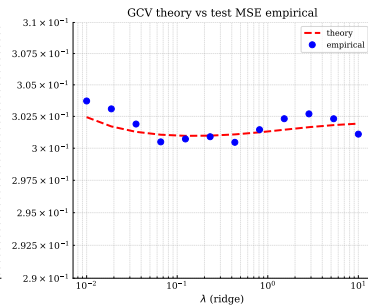
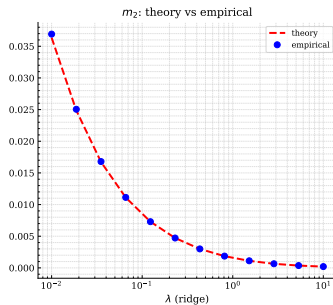
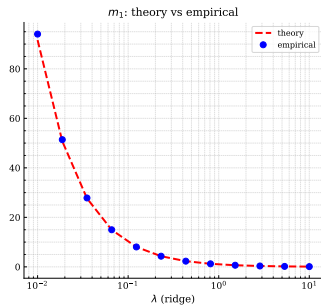
**Power-law resolvents:** All three trace resolvents  $m_1, m_2, m_\beta$  plotted against ridge  $\lambda$  (middle).  $m_\beta$  shows divergence from  $m_2$ . Theoretical curves for all three trace resolvents track empirical plots well. Right figure depicts theoretical GCV prediction and with empirical test MSE. Plots generated with  $n = 2250, N = 3000, p = 1500, \sigma_\varepsilon^2 = 0.3, \sigma_W^2 = 1.0, \alpha = 0.5, r = 0.8$  and error bars over 20 trials

# Experiments



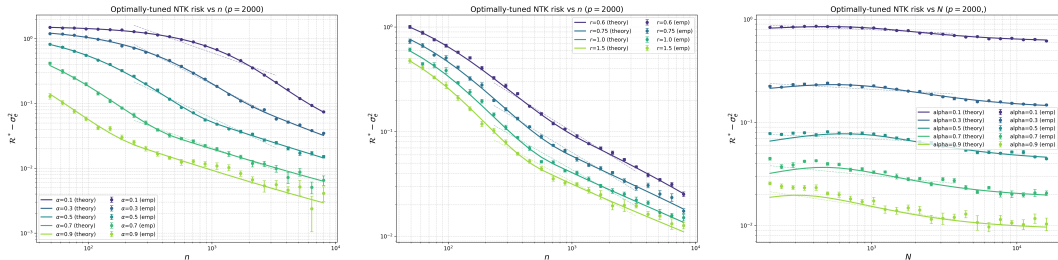
**Theory vs empirical  $\alpha = 0.5$ :** lots for  $m_1$  (left),  $m_2$  (middle), and  $\text{GCV}_\lambda$  (right) against ridge  $\lambda$ . All three show plots strong agreement across. Plots generated with  $n = 2400$ ,  $N = 1800$ ,  $p = 1200$ ,  $\sigma_\varepsilon^2 = 0.3$ ,  $\sigma_W^2 = 1.0$  and isotropic random labels.

# Experiments



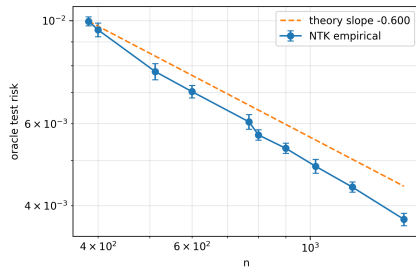
Theory vs empirical  $\alpha = 1.5$ : Same setup as figure on preceding slide, with larger capacity exponent  $\alpha > 1$ .  
Test loss (right) is noisier, but still tracked by theoretical curve.

# Experiments



**Optimally-tuned Risk:** Excess NTK risk of optimally-tuned ridge  $\lambda = \lambda^*$  (markers with  $\pm$ SE over 10 trials) on log-log axes. Theoretical curves plotted in solid lines, showing close alignment across all regimes. **Left:** excess risk vs training samples  $n$  with varying capacity exponent  $\alpha$  and fixed  $N = 5000, r = 1.0$ . As  $\alpha$  increases, curves straighten out to follow power-law decay. **Middle:** Varying values of source exponent  $r$ . **Right:** risk against width  $N$  for sweeping  $\alpha$ , and fixed  $n = 2000, r = 0.7$ . Throughout  $\sigma_W = 1, \sigma_\varepsilon^2 = 0.01, p = 2000$ .

**Scaling law of NTK regression** Under a further assumption on the GCV- NTK Risk deterministic equivalence, we derive a scaling law for finite-width NTK. The simulation uses a **capacity** strength of  $\alpha = 1.5$  and a **source** complexity of  $r = 0.5$ . The resulting empirical risk exponent of  $\theta_{\text{NTK}} \approx 0.687$  aligns closely with the theoretical prediction of 0.6, confirming that the Gaussian-equivalent model accurately captures the scaling behavior of finite-width neural networks in this regime. Moreover, classical KRR *source-capacity* scaling exponents are recovered in the  $N \rightarrow \infty$  limit.



**Scaling Law:** Test risk scaling of a finite-width Gaussian-equivalent NTK on unnormalized power-law data ( $\alpha = 1.5, r = 0.5$ ).

# Chapter 3: Eigen-Spike Emergence and Quadratic Equivalents for Conjugate Kernels on Nonlinearly Separable Data

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# Conjugate Kernel (CK)

Define the **Conjugate Kernel (CK)**:

## Conjugate Kernel

$$\mathbf{K}^{\text{CK}} := \mathbf{Y}^\top \mathbf{Y}, \quad \mathbf{Y} = \frac{1}{\sqrt{N}} \sigma(\mathbf{W}\mathbf{X})$$

**Notation:**

$$\mathbf{X} \in \mathbb{R}^{d \times n}, \quad \mathbf{W} \in \mathbb{R}^{N \times d}, \quad \mathbf{W}_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

**Motivations:**

- CK captures the *similarity* structure of data in random feature space of NN.
- **Spectral properties** of CK provide insight into
  - **Representation quality**: CK eigenvalues describe strength of learned similarities.
  - **Feature extraction**: CK Eigenvectors indicate direction of task-model alignment.
  - **Generalization**: CK spectra dictates generalization ability of NTK and RF models.

## Motivating Question:

What are the machine learning implications of the Linear Equivalence (LE):

$$\sigma(\mathbf{WX}) \approx b_1 \mathbf{WX} + b_2 \Theta, \quad \Theta_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

LE says CK behaves as a **linear kernel** plus isotropic noise

## Problem

The LE **hides important nonlinear task structure**

- For **nonlinearly separable data**, LE implies spectral methods on CK fail.
- **Neural Networks**, in contrast, have been shown to learn complex nonlinear structure in data
- LE fails to explain *how* or *when* nonlinearity  $\sigma$  helps learning task.

## Motivating Question:

What are the machine learning implications of the Linear Equivalence (LE):

$$\sigma(\mathbf{WX}) \approx b_1 \mathbf{WX} + b_2 \Theta, \quad \Theta_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

To answer this, we consider the central spectral question of this chapter.

## Central Spectral Question

When can a neural network transfer a **nonlinear pattern** in data to a **linearly accessible direction**<sup>4</sup> that allows for linear classification?

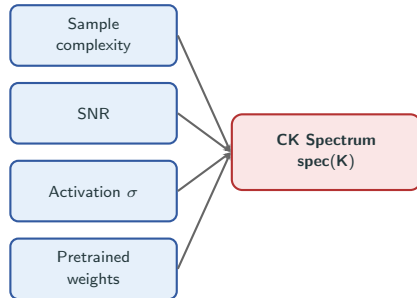
**Key mechanism:** BBP (Baik–Ben Arous–Péché) transitions [Baik-Ben Arous-Péché, 2005]– when an informative outlier eigenvalue separates from the bulk spectrum, and its corresponding eigenvector aligns with the signal.

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<sup>4</sup>aligned eigen-spike

**Practical Implications:** ML practitioners tune various “knobs” to improve model performance:

- **Sample complexity** :  $n$  vs. feature dimension  $d$  vs. width  $N$
- **Signal-to-noise ratio (SNR)**: ratio of meaningful signal to random noise.
- **Activation function**: Choice of nonlinearity  $\sigma$
- **Training**: Random initialization vs. pretrained weights  $\mathbf{W}$



## Question:

How do each of these knobs affect the spectral structure of  $\mathbf{K}^{\text{CK}}$ , and in turn the *learnability* of nonlinear patterns?

# XOR Gaussian Mixture Dataset

## Four-Component Gaussian Mixture Model

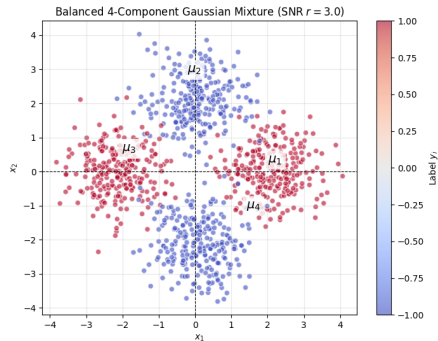
(GMM):  $\mathbf{x}_i \stackrel{\text{i.i.d.}}{\sim} \frac{1}{4} \sum_{a=1}^4 \mathcal{N}\left(\frac{1}{\sqrt{d}} \boldsymbol{\mu}_a, \frac{1}{d} \mathbf{I}_d\right)$

Cluster means: for SNR parameter  $r \geq 0$ ,

$$\boldsymbol{\mu}_1 = r \mathbf{u}_1, \quad \boldsymbol{\mu}_2 = r \mathbf{u}_2, \quad \boldsymbol{\mu}_3 = -\boldsymbol{\mu}_1, \quad \boldsymbol{\mu}_4 = -\boldsymbol{\mu}_2$$

with  $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^d$  orthonormal.

**Labels:**  $y_i = +1$  for clusters  $\{1, 3\}$ ,  $y_i = -1$  for clusters  $\{2, 4\}$ , making the classification problem **nonlinearly separable**.



## Key Property: Nonlinear Separability

- **Not linearly separable:** class means cancel  $\Rightarrow$  no linear method works
- **Quadratically separable:** second-order features (e.g.,  $\mathbf{x}_i^\top \mathbf{A} \mathbf{x}_i$ ) can separate

## Why study XOR?

- **Canonical nonlinearly separable dataset:** widely studied in Deep Learning theory (Refinetti et al. 2021, Glasgow et al. 2023)
- **Quadratic feature channel:** isolates quadratic features, which can generalize to more complex nonlinear tasks.

## Proxy for nonlinear learnability

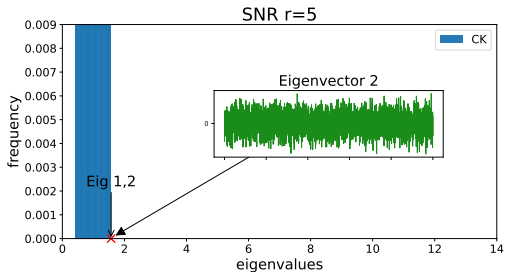
When do **spike eigenvalues** emerge from the bulk of  $\mathbf{K}^{\text{CK}}$  whose eigenvectors **asymptotically align** with XOR-labels?

When yes  $\Rightarrow$  linear spectral classification via  $\mathbf{K}^{\text{CK}}$  is possible.

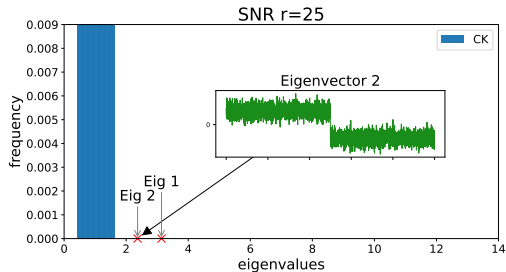
When no  $\Rightarrow$  nonlinear structure remains hidden from linear readouts.

# Example: SNR Phase Transition

## Phase Transition in SNR



$r = 5$ : no eigenvalues separate from bulk



$r = 25$ : two eigenvalues **emerge**

## Four Practical “Knobs”

### Knob 1:

Sample complexity  
 $n$  vs  $d$  vs  $N$

### Knob 2:

Signal-to-noise  
ratio (SNR)

### Knob 3:

Pretrained/spiked  
weights  $\mathbf{W}$

### Knob 4:

Activation  
choice  $\sigma$

For each configuration of these knobs, we ask:

1. Do **outlier eigenvalues** emerge in the spectrum of  $\mathbf{K}^{\text{NTK}}$ ?
2. Do their **eigenvectors align with XOR labels  $\mathbf{y}$** ?
3. Can we perform **linear spectral classification**?

We identify **BBP-type phase transitions** for each knob, and consider their practical implications.

Write XOR dataset as a signal-plus-noise **low rank perturbation**:

$$\mathbf{X} = \mathbf{Z} + \mathbf{M}, \quad \mathbf{Z}_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

where  $\mathbf{M} = r\sqrt{\frac{n}{2d}}(\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{u}_2\mathbf{v}_2^\top)$  is a **rank-2** signal, and  $\mathbf{v}_1 \perp \mathbf{v}_2 \in \mathbb{R}^n$  encode cluster membership, and  $\text{SNR} := r^2$

### Assumptions

The activation function  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  satisfies

- $\sigma$  is centered and normalized with respect to the Gaussian:  
 $\mathbb{E}[\sigma(\xi)] = 0, \mathbb{E}[\sigma(\xi)^2] = 1.$
- $\sigma$  is three-times differentiable with bounded derivatives.

**Hermite Coefficients of  $\sigma$ :**

$$b_\sigma := \mathbb{E}[\sigma'(\xi)] = \mathbb{E}[\xi \sigma(\xi)]$$

$$c_\sigma := \mathbb{E}[\sigma''(\xi)] = \mathbb{E}[(\xi^2 - 1)\sigma(\xi)]$$

Write XOR dataset as a signal-plus-noise **low rank perturbation**:

$$\mathbf{X} = \mathbf{Z} + \mathbf{M}, \quad \mathbf{Z}_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

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- $\sigma$  is centered and normalized with respect to the Gaussian:  
 $\mathbb{E}[\sigma(\xi)] = 0, \mathbb{E}[\sigma(\xi)^2] = 1.$
- $\sigma$  is three-times differentiable with bounded derivatives.

### Main Takeaway:

- $b_\sigma$ : controls the **linear channel**
- $c_\sigma$ : controls the **quadratic channel**  $\longleftarrow$  **key for XOR!**

## Summary of Main Results

| Regime                         | SNR               | Sample         | Weights | Label Align? |
|--------------------------------|-------------------|----------------|---------|--------------|
| Case 1 (Baseline)              | $\Theta(1)$       | $n \asymp d$   | Random  | No           |
| Case 2 (Large SNR)             | $\Theta(n^{1/4})$ | $n \asymp d$   | Random  | Yes          |
| Case 3 (Trained $\mathbf{W}$ ) | $\Theta(1)$       | $n \asymp d$   | Spiked  | Yes          |
| Case 4 (Quad. size)            | $\Theta(1)$       | $n \asymp d^2$ | Random  | Yes          |

**Common thread:** In each successful case, the **quadratic Hermite component**  $c_\sigma$  creates a feature channel that makes the nonlinear XOR structure spectrally visible.

**Activation requirement:**  $c_\sigma \neq 0$  is **necessary** in all cases. If  $c_\sigma = 0$  (e.g.,  $\tanh$  after centering/normalization), linear classification via CK is impossible.

## Outline of Approach

**Core idea:** Expand  $\sigma(\mathbf{WX})$  around **null-model**  $\sigma(\mathbf{WZ})$  and retain **second-order Hermite coefficients**.

### Proposition: Quadratic Equivalent (QE)

Let  $\mathbf{Y}_0 = \frac{1}{\sqrt{N}}\sigma(\mathbf{WZ})$ ,  $\mathbf{g}_k = \mathbf{W}\mathbf{u}_k$ ,  $\theta_{\text{SNR}} = r\sqrt{n/(2d)}$ . Then:

$$\mathbf{Y}_{\text{QE}} := \underbrace{\mathbf{Y}_0}_{\text{bulk}} + \underbrace{\frac{\theta_{\text{SNR}} b_\sigma}{\sqrt{N}} (\mathbf{g}_1 \mathbf{v}_1^\top + \mathbf{g}_2 \mathbf{v}_2^\top)}_{T_1 \text{ (linear spikes)}} + \underbrace{\frac{\theta_{\text{SNR}}^2 c_\sigma}{\sqrt{N}} (\mathbf{g}_1^{\odot 2} \mathbf{v}_1^{\odot 2\top} + \mathbf{g}_2^{\odot 2} \mathbf{v}_2^{\odot 2\top})}_{T_2 \text{ (quadratic spikes)}}$$

satisfies  $\|\mathbf{Y} - \mathbf{Y}_{\text{QE}}\| \rightarrow 0$  in probability.

- $\mathbf{Y}_0$ : determines bulk spectrum  $\mu$  (may also create uninformative outliers)
- $T_1 \sim \theta_{\text{SNR}} b_\sigma$ : linear spikes, eigenvectors align with  $\mathbf{v}_1, \mathbf{v}_2$  but  $\perp \mathbf{y}$
- $T_2 \sim \theta_{\text{SNR}}^2 c_\sigma / \sqrt{n}$ : quadratic spikes, can align with  $\mathbf{y}$  **appears when  $r = \Theta(n^{1/4})$**

**Proportional Regime:**  $n, d, N \rightarrow \infty$

$$n/N \rightarrow \varphi \in (0, \infty), \quad n/d \rightarrow \psi \in (0, \infty), \quad r = \Theta(1)$$

**Bulk law:** Empirical spectral distribution (ESD) of  $\mathbf{K}^{\text{CK}}$  converges to

$$\mu := \rho_{\varphi}^{\text{MP}} \boxtimes \nu, \quad \nu = (1 - b_{\sigma}^2) \delta_0 + b_{\sigma}^2 \rho_{\psi}^{\text{MP}}$$

(free multiplicative convolution of Marchenko–Pastur laws) [FW20].

**Key transforms:**

$$z(s) = -\frac{1}{s} + \phi \int \frac{x}{1 + xs} \nu(dx),$$

$$\varphi(s) = -\frac{sz'(s)}{z(s)},$$

$$T(s) = \frac{z(s)s^2 - (\phi - 1)s}{\phi},$$

$$T^{-1}(t) = -\frac{t(1 - \psi b_{\sigma}^2 t)}{1 + (1 - \psi b_{\sigma}^2)t - \psi b_{\sigma}^2(1 - b_{\sigma}^2)t^2},$$

### Theorem (Finite-SNR, Proportional Limit)

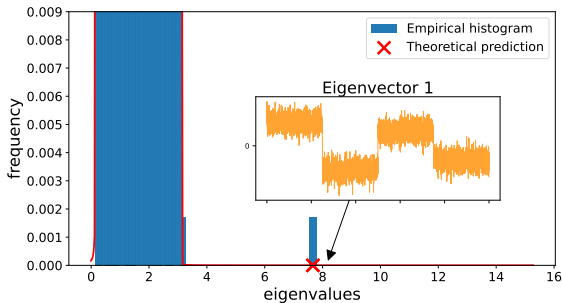
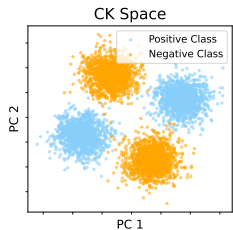
Under the above setting with  $r = \Theta(1)$ :

- (i) **Bulk:** *ESD of  $\mathbf{K}^{\text{CK}} \rightarrow \mu$ .*
- (ii) **BBP for covariance spike:** *If  $\tau > \tau_{\text{crit}}$  and  $z'(-1/\Lambda_\tau) > 0$ , then  $\mathbf{K}^{\text{CK}}$  has an outlier at  $\hat{\lambda}_\tau = z(-1/\Lambda_\tau) + o_{n,\mathbb{P}}(1)$ . This spike is **uninformative** (present even for  $r = 0$ ).*
- (iii) **BBP for mean spike:** *Similar BBP transition for the mean direction.*
- (iv) **Two linear spikes from XOR:** *If  $z'(s_{\text{lin}}) > 0$ , two outlier eigenvalues at  $z(s_{\text{lin}})$ , with eigenvectors aligning with  $\mathbf{v}_1, \mathbf{v}_2$  (within-class structure).*
- (v) **No label alignment:** **Every** outlier eigenvector of  $\mathbf{K}^{\text{CK}}$  is asymptotically orthogonal to  $\mathbf{y}$ :

$$\|\hat{\mathbf{V}} \mathbf{y}\|^2 \rightarrow 0 \quad \text{a.s.}$$

## Case 1: Failure Case

**Failure case:** Finite-SNR proportional regime is the baseline where **linear classification** fails.



### Observations:

- XOR visible in PC space **but not linearly separable**
- Theory (red) matches
- Leading eigenvec  $\perp \mathbf{y}$

$n=5000$ ,  $N=d=150000$ ,

$r=6$ ,  $\sigma \propto \text{ReLU}$

### Key Takeaway

The **linear spikes** (from  $b_\sigma$ ) carry within-class information  $(\mathbf{v}_1, \mathbf{v}_2)$ , but  $\mathbf{y} \perp \{\mathbf{v}_1, \mathbf{v}_2\}$ .

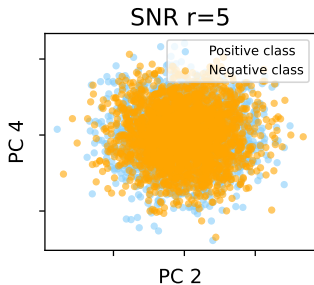
In the proportional regime with finite SNR, the CK exhibits outliers, but **linear classification of XOR fails**. The LE correctly predicts this failure.

**Additional negative result:** Kernel spectral clustering [Couillet-Florent Benaych-Georges 2016] (normalized Laplacian with Euclidean distance kernel) **also fails** for XOR in this regime.

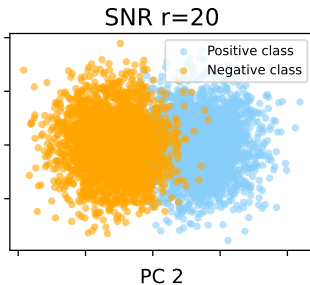
**Remark:** The uninformative spike can be removed if  $\tau \leq \tau_{\text{crit}}$ , i.e.,  $\sqrt{\psi}(\frac{c_\sigma^2}{2} - b_\sigma^2) \leq b_\sigma^2$ . If  $c_\sigma/\sqrt{2}$  is small relative to  $b_\sigma$  (e.g., centered normalized GELU).

## Knob: Increase SNR — PCA Phase Transition

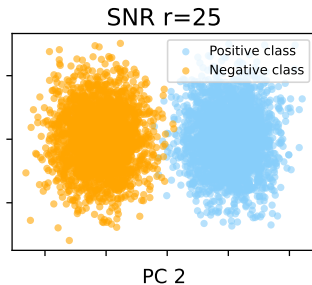
**Phase Transition in Principal Component (PC):** PC 2 vs PC 4 of CK over increasing SNR. Yellow:  $-1$  labels, Blue:  $+1$  labels.



$r = 5$  (no separation)



$r = 20$  (transition)



$r = 25$  (linearly separable)

**Clear phase transition:** Around  $r = 20$ – $25$ , XOR becomes **linearly separable** in CK principal component space.

## Case 2: Setup — Large SNR

**Idea:** Increase the SNR so that the **quadratic Hermite component** becomes non-negligible.

**Scaling:** Same proportional limit, but now

$$r = r_0 n^{1/4}, \quad r_0 \in (0, \infty) \text{ fixed.}$$

**New constants:**

$$\kappa_0 = \frac{r_0^4 \psi^2}{4}, \quad \beta_{\text{quad}} = c_\sigma^2 \kappa_0, \quad \tilde{\beta}_{\text{quad}} = \beta_{\text{quad}} + \tau, \quad s_{\text{quad}} = T^{-1}(\beta_{\text{quad}}).$$

**Key observation:** With  $\theta_{\text{SNR}} = r\sqrt{n/(2d)} = \Theta(n^{1/4})$ , the quadratic term

$$T_2 = \frac{\theta_{\text{SNR}}^2 c_\sigma}{2\sqrt{N}} \left( \mathbf{g}_1^{\odot 2} \mathbf{v}_1^{\odot 2\top} + \mathbf{g}_2^{\odot 2} \mathbf{v}_2^{\odot 2\top} \right)$$

becomes  $O(1)$  and **comparable to the bulk**.

## Case 2: Emergence of Quadratic Informative Spike for Large-SNR Proportional Limit

### Theorem (Large-SNR Proportional Limit)

Under  $r = r_0 n^{1/4}$ , let  $\mathbf{P}_{\text{lin}} = \mathbf{v}_1 \mathbf{v}_1^\top + \mathbf{v}_2 \mathbf{v}_2^\top$  and  $\mathbf{K}_{\perp}^{\text{CK}} = \mathbf{P}_{\perp} \mathbf{K}^{\text{CK}} \mathbf{P}_{\perp}$ :

- (i) **Bulk:** ESD of  $\mathbf{K}^{\text{CK}} \rightarrow \mu$ .
- (ii) **Two diverging linear outliers:** If  $b_{\sigma} \neq 0$ , top two eigenvalues diverge as  $\hat{\lambda}_{1,2} = b_{\sigma}^2 r_0^2 \psi \cdot n^{1/2}(1 + o_P(1))$ , but their eigenvectors are  $\perp \mathbf{y}$ .
- (iii) **Quadratic  $1_n$ -spike:** BBP transition for  $1_n$ -direction at effective strength  $\tilde{\beta}_{\text{quad}}$ .
- (iv) **Quadratic  $\mathbf{y}$ -spike (label alignment!):** If  $z'(s_{\text{quad}}) > 0$ , then  $\mathbf{K}_{\perp}^{\text{CK}}$  has an outlier whose eigenvector satisfies:

$$\left[ \frac{1}{n} \left| \hat{\mathbf{v}}_{\mathbf{y}}^\top \mathbf{y} \right|^2 \xrightarrow{P} \frac{s_{\text{quad}} z'(s_{\text{quad}})}{\beta_{\text{quad}} T'(s_{\text{quad}})} > 0 \right]$$

### Key Takeaway

By increasing SNR to  $r = \Theta(n^{1/4})$ , the quadratic nonlinearity creates **label-informative outliers**  $\Rightarrow$  **linear spectral classification of XOR becomes possible!**

**Why does this work?** The identity

$$\mathbf{v}_1^{\odot 2} = \frac{1}{\sqrt{n}}(1_n + \mathbf{y}), \quad \mathbf{v}_2^{\odot 2} = \frac{1}{\sqrt{n}}(1_n - \mathbf{y})$$

shows that **element-wise squaring** converts within-class directions  $\mathbf{v}_1, \mathbf{v}_2$  into label  $\mathbf{y}$ .

**Role of activation choice:**

- If  $b_\sigma = 0$ : diverging linear outliers vanish  $\Rightarrow$  **simpler** spectral classification
- If  $c_\sigma = 0$ : quadratic channel absent  $\Rightarrow$  **no label alignment possible**
- The activation  $\sigma$  must have  $c_\sigma \neq 0$  for nonlinear classification!

## Conclusion

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## Conclusion:

- In Chapter 2, we utilized a universality result (**Gaussian Equivalence**) to study finite-width NTK regression under a dual-power law model.
  - We introduced a novel technique to derive deterministic equivalents for trace resolvents that characterize the prediction risk of NTK, extending previous results beyond isotropic setting.
  - As an application, we derived a *source-capacity* scaling law for finite-width NTK, contributing to the fast growing literature on **neural scaling** laws in kernel methods.
- In Chapter 3, we study the practical implications of the **Gaussian Equivalence** using spike eigenvalues of the CK and their task-alignment as a proxy for nonlinear learnability.
  - We develop a **Quadratic Equivalent (QE)** framework that allows for the precise **BBP**-type phase transitions in four regimes of practical interest.

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**Thank you!**

# Appendix

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## Chapter 2 Appendix

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Decompose Jacobian over  $\theta$  into first and second layer weights,

$$\mathbf{K}^{\text{NTK}}(\mathbf{x}, \mathbf{x}') = \underbrace{\nabla_{\mathbf{w}_1} f_{\theta}(\mathbf{x}; \theta_0)^{\top} \nabla_{\mathbf{w}_1} f_{\theta}(\mathbf{x}'; \theta_0)}_{\mathbf{K}_1(\mathbf{x}, \mathbf{x}')} + \underbrace{\nabla_{\mathbf{w}} f_{\theta}(\mathbf{x}; \theta_0)^{\top} \nabla_{\mathbf{w}} f_{\theta}(\mathbf{x}'; \theta_0)}_{\mathbf{K}_2(\mathbf{x}, \mathbf{x}')}$$

- Define the second layer Kernel  $\mathbf{K}_2(\mathbf{X}, \mathbf{X}) = \mathbf{F}^{\top} \mathbf{F}$ , where  $\mathbf{F} = \frac{1}{\sqrt{N}} \sigma(\mathbf{W}_1 \mathbf{X})$ .
- A quick computation gives the first layer kernel<sup>5</sup>  
 $\mathbf{K}_1(\mathbf{X}, \mathbf{X}) = \frac{\mathbf{X}^{\top} \mathbf{X}}{p} \odot \frac{(\mathbf{F}')^{\top} \text{diag}(\mathbf{w})^2 \mathbf{F}'}{N}$ , where  $\mathbf{F}' = \frac{1}{\sqrt{N}} \sigma'(\mathbf{W}_1 \mathbf{X})$ .
  - $\mathbf{K}_2$  is also called the **Conjugate Kernel (CK)**.
  - As  $\sigma_W \rightarrow 0$ ,  $\mathbf{K}^{\text{NTK}}$  degenerates to the CK and the learning problem reduces to Random Feature (RF) regression [RR07].

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<sup>5</sup> $\odot$  denotes Hadamard (entry-wise) product

**Standard tool** from RMT in proportional asymptotic ( $n \asymp N \asymp p$ ) analysis of nonlinear kernel models.

## Linear Equivalence (LE)<sup>6</sup>

$$\sigma(\mathbf{W}_1 \mathbf{X}) \approx b_1 \mathbf{W}_1 \mathbf{X} + b_2 \Theta, \quad \Theta_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

when  $\mathbb{E}[\sigma(g)] = 0$ , where  $b_1 = \mathbb{E}[\sigma'(g)]$ ,  $b_2^2 = \mathbb{E}[\sigma(g)^2] - \tau^2 b_1^2$  and  $g \sim \mathcal{N}(0, \tau^2)$ .

- Under LE, second layer Kernel simplifies  $\mathbf{K}_2 \approx \mathbf{F}_{\text{lin}}^\top \mathbf{F}_{\text{lin}}$ , where  $\mathbf{F}_{\text{lin}} = \frac{1}{\sqrt{N}}(b_1 \mathbf{W}_1 \mathbf{X} + b_2 \Theta)$ .
- By centering and controlling the off-diagonal terms of  $\mathbf{K}_1$ , we have

$$\mathbf{K}_1 \approx b_1^2 \sigma_W^2 p^{-1} \mathbf{X}^\top \mathbf{X} + \tau^2 \sigma_W^2 (c_1 - b_1^2) \mathbf{I}_n, \quad c_1 := \mathbb{E}[\sigma'(g)^2]$$

**Key references:** Mei–Montanari (2022), Hu–Lu (2022), Goldt et al. (2021), Adlam–Pennington (2020)

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<sup>6</sup>approximation  $\approx$  holds on order of bulk statistics

**Standard tool** from RMT in proportional asymptotic ( $n \asymp N \asymp p$ ) analysis of nonlinear kernel models.

### Linear Equivalence (LE)

$$\sigma(\mathbf{W}_1 \mathbf{X}) \approx b_1 \mathbf{W}_1 \mathbf{X} + b_2 \Theta, \quad \Theta_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

when  $\mathbb{E}[\sigma(g)] = 0$ , where  $b_1 = \mathbb{E}[\sigma'(g)]$ ,  $b_2^2 = \mathbb{E}[\sigma(g)^2] - \tau^2 b_1^2$  and  $g \sim \mathcal{N}(0, \tau^2)$ .

— Collecting both substitutions, we define the *linearized NTK*

$$\mathbf{K}_{\text{lin}} = \mathbf{F}_{\text{lin}}^\top \mathbf{F}_{\text{lin}} + \sigma_W^2 \left( p^{-1} b_1^2 \mathbf{X}^\top \mathbf{X} + \tau^2 (c_1 - b_1^2) \mathbf{I}_n \right) + z \mathbf{I}_n, \quad \mathbf{F}_{\text{lin}} = \frac{1}{\sqrt{N}} (b_1 \mathbf{W}_1 \mathbf{X} + b_2 \Theta)$$

We will drop the *lin* subscript, and study KRR on the *linearized* kernel  $\mathbf{K}$  directly.

**Universality Principle:** Training and test error of kernel ridge regression on  $\mathbf{K}^{\text{NTK}}$  are asymptotically equivalent to that of regression on *linearized* kernel  $\mathbf{K}$ .

### Gaussian Equivalence Theorem (GET)

The following holds pointwise for each fixed  $\lambda \geq 0$ .

$$|\mathcal{R}(\lambda) - \mathcal{R}_{\text{GE}}(\lambda)| = o_{n,\mathbb{P}}(1)$$

- $\mathcal{R}_{\text{GE}}(\lambda)$  denotes prediction risk of KRR on the linearized NTK kernel.
- **Conjecture** GET extends to power-law input (A0).

**Key references:** Adlam et al. (2019), Hu–Lu (2022), Montanari–Saeed (2022), Peche et al. (2019)

## NTK Scaling Law

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## Assumption 1: Unnormalized power-law

Replace Lecun factor  $p^{-1/2}$  on sample covariance with trace-normalization

$T_p = \text{Tr } \Sigma = \sum_{j=1}^p j^{-\alpha}$ , where

$$\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p), \quad \sigma_j = j^{-\alpha}, \quad \alpha > 1.$$

Normalize signal vector  $\beta_j = c_{\alpha,r} j^{-r}$ , where  $c_{\alpha,r}$  is chosen to ensure  $\beta^\top \Sigma \beta = 1$ , and assume  $\alpha + 2r > 1$ .

## Remarks

- Trace-normalization ensures pre-activations are non-degenerate as  $p \rightarrow \infty$ . A similar normalization convention was taken in Loureiro et al (2025).
- Condition  $\alpha + 2r > 1$  ensures  $\|f^*\|_{L^2} < \infty$  as  $p \rightarrow \infty$ .
- For simplicity, we assume noiseless test label  $y(\mathbf{x}) = \mathbf{x}^\top \beta$ , although this can be dropped by subtracting the variance of test noise from GCV estimator.

## Scaling Law: Assumptions

Define constants

$$\bar{\sigma}_{p,j} := \frac{\sigma_j}{T_p}, \quad c_{\text{NTK}} := b_1^2(1 + \sigma_W^2) > 0,$$

For effective ridge  $\mu > 0$ , define **effective dimension** and **source-bias** profiles

$$\mathcal{N}_p(\mu) := \sum_{j=1}^p \frac{c_{\text{NTK}} \bar{\sigma}_{p,j}}{c_{\text{NTK}} \bar{\sigma}_{p,j} + \mu}, \quad \mathcal{B}_p(\mu) := \sum_{j=1}^p \sigma_j \beta_{p,j}^2 \left( \frac{\mu}{c_{\text{NTK}} \bar{\sigma}_{p,j} + \mu} \right)^2.$$

### Assumption 2: Local GCV Deterministic Equivalent

Let  $\mu := \lambda + \lambda_{\text{diag}}$  be the effective ridge appearing in the HJ deterministic equivalent, and assume either  $\lambda_{\text{diag}} = 0$  or  $\lambda_{\text{diag}} = o(\mu_{\lambda^*})$ . Further assume there exists constants  $C_B, C_V \in (0, \infty)$  such that uniformly for  $\mu_\lambda \downarrow 0$  with  $\mathcal{N}_p(\mu) = o(n \wedge N)$  and  $\mu_\lambda^{-1/\alpha} = o(p)$ ,

$$R_{n,N,p}(\lambda) = C_B \mathcal{B}_p(\mu_\lambda) + C_V \left( \frac{1}{n} + \frac{1}{N} \right) \mathcal{N}_p(\mu_\lambda) + o \left( \mathcal{B}_p(\mu_\lambda) + \left( \frac{1}{n} + \frac{1}{N} \right) \mathcal{N}_p(\mu_\lambda) \right).$$

## Theorem (Finite-width NTK Scaling Law)

Under Assumptions 1 and 2, define

$$q_{\alpha,r} := \min\left\{2, \frac{\alpha + 2r - 1}{\alpha}\right\}$$

Suppose  $\alpha + 2r \neq 2\alpha + 1$ , and  $\frac{1}{n} + \frac{1}{N} \rightarrow 0$  with  $p$  sufficiently large, then the optimally tuned NTK risk satisfies the following **scaling law**

$$\inf_{\lambda \geq 0} \mathcal{R}_{n,N,p}(\lambda) \asymp \left(\frac{1}{n} + \frac{1}{N}\right)^\theta, \quad \theta := \frac{\alpha q_{\alpha,r}}{\alpha q_{\alpha,r} + 1} < 1$$

Moreover, the **optimal ridge** satisfies  $\mu_{\lambda^*} \asymp \left(\frac{1}{n} + \frac{1}{N}\right)^{\frac{\alpha}{\alpha q_{\alpha,r} + 1}}$

Recover classical KRR *source-capacity* exponent Caponnetto–De Vito (2007) in the  $N \rightarrow \infty$  limit, under the renormalization of source parameter  $r_{\text{CdV}} = \frac{q}{2}$ .

## Two Regimes

**Non-saturated source regime:** if  $\alpha + 2r < 2\alpha + 1$ , then  $q_{\alpha,r} = \frac{\alpha+2r-1}{\alpha}$  and

$$\inf_{\lambda \geq 0} \mathcal{R}_{n,N,p}(\lambda) \asymp \left( \frac{1}{n} + \frac{1}{N} \right)^{\frac{\alpha+2r-1}{\alpha+2r}}, \quad \mu_{\lambda^*} \asymp \left( \frac{1}{n} + \frac{1}{N} \right)^{\frac{\alpha}{\alpha+2r}}$$

**Saturated source regime:**  $\alpha + 2r > 2\alpha + 1$ , then  $q_{\alpha,r} = 2$

$$\inf_{\lambda \geq 0} R_{n,N,p}(\lambda) \asymp \left( \frac{1}{n} + \frac{1}{N} \right)^{\frac{2\alpha}{2\alpha+1}}, \quad \mu_{\lambda_*} \asymp \left( \frac{1}{n} + \frac{1}{N} \right)^{\frac{\alpha}{2\alpha+1}}.$$

## Remarks:

- **Non-saturated source :**  $r < \frac{\alpha+1}{2}$  rate depends on both  $(\alpha, r)$ .
- **Saturated source:**  $r > \frac{\alpha+1}{2}$  rate set by capacity  $\alpha$  only, source parameter  $r$  no longer contributes.

### Corollary

Under the setup of the previous Theorem, if  $N \asymp n^\ell$  for some  $\ell > 0$ , then

$$\left(\frac{1}{n} + \frac{1}{N}\right) \asymp n^{-\min(1,\ell)}.$$

Consequently,

$$\inf_{\lambda \geq 0} R_{n,N,p}(\lambda) \asymp n^{-\theta_{\text{NTK}}(\alpha,r,\ell)}, \quad \theta_{\text{NTK}}(\alpha,r,\ell) := \min\{1,\ell\} \frac{\alpha q_{\alpha,r}}{\alpha q_{\alpha,r} + 1}$$

**Data vs Width Bottleneck:** When  $\ell < 1$ , width is the limiting factor, and  $\theta_{\text{NTK}}$  is degraded by the same factor. When  $\ell > 1$ , data binds and finite-width NTK gets full kernel exponent  $\theta_{\text{NTK}}$ .

# Scaling Law: Proof Idea

**Core idea:** Split modes at **resolution-scale**  $J_\mu := \lfloor \mu^{-1/\alpha} \rfloor$ .

$$\frac{c_{\text{NTK}} \bar{\sigma}_{p,j}}{c_{\text{NTK}} \bar{\sigma}_{p,j} + \mu} \asymp \begin{cases} 1, & j \leq J_\mu \\ j^{-\alpha}/\mu, & j > J_\mu \end{cases}, \quad \frac{\mu}{c_{\text{NTK}} \bar{\sigma}_{p,j} + \mu} \asymp \begin{cases} \mu j^\alpha, & j \leq J_\mu \\ 1, & j > J_\mu \end{cases}$$

**Step one:** Estimate *effective dimension*  $\mathcal{N}_p(\mu)$ .

$$\mathcal{N}_p(\mu) \asymp \sum_{j \leq J_\mu} 1 + \mu^{-1} \sum_{J_\mu < j \leq p} j^{-\alpha} \asymp \boxed{\mu^{-1/\alpha}}.$$

**Step two:** Estimate the *source-bias profile*  $\mathcal{B}_p(\mu) = B_{\leq}(\mu) + B_{\geq}(\mu)$ , where

$$B_{\leq}(\mu) \asymp \mu^2 \sum_{j \leq J_\mu} j^{\alpha-2r}, \quad B_{\geq}(\mu) \asymp \sum_{j > J_\mu} j^{-(\alpha+2r)}$$

When  $\alpha + 2r > 2\alpha + 1$ , the below-resolution sum is  $O(1)$  and  $\mathcal{B}_p(\mu) \asymp \mu^2$ . Otherwise,  $\mathcal{B}_p(\mu) \asymp \mu^{(\alpha+2r-1)/\alpha}$ . In either case,

$$\boxed{\mathcal{B}_p(\mu) \asymp \mu^{q_{\alpha,r}}}$$

**Step three:** Optimize the deterministic equivalent.

$$\mathcal{R}_{n,N,p}(\lambda) \asymp \mu_\lambda^{q_{\alpha,r}} + \left(\frac{1}{n} + \frac{1}{N}\right) \mu_\lambda^{-1/\alpha}$$

Balance the two terms when  $\mu_{\lambda^*} \asymp \left(\frac{1}{n} + \frac{1}{N}\right) \mu_\lambda^{-1/\alpha}$ . At this optimal ridge,

$$\inf_{\lambda \geq 0} \mathcal{R}_{n,N,p}(\lambda) \asymp \left(\frac{1}{n} + \frac{1}{N}\right)^{\frac{\alpha q_{\alpha,r}}{\alpha q_{\alpha,r} + 1}}$$

**Remark:** At **logarithmic boundary**  $\alpha + 2r = 2\alpha + 1$ ,

$$\inf_{\lambda \geq 0} \mathcal{R}_{n,N,p}(\lambda) \asymp \left(\frac{1}{n} + \frac{1}{N}\right)^{\frac{2\alpha}{2\alpha+1}} \left(\log \frac{1}{n^{-1} + N^{-1}}\right)^{\frac{1}{2\alpha+1}}$$

## Chapter 3 Appendix

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## Case 3: Setup — Spiked Weights

**Motivation:** Pretrained NNs have weight matrices with spectral spikes (heavy-tailed ESDs observed empirically). Model this as a **rank-one perturbation**:

$$\mathbf{W}_1 = \mathbf{W} + \theta \mathbf{a} \mathbf{b}^\top, \quad \theta = \theta_0 N^{1/4}$$

- $\mathbf{W}$ : random initialization ( $W_{ij} \sim \mathcal{N}(0, 1)$ )
- $\mathbf{a} \sim \mathcal{N}(0, I_N/N)$ : random direction
- $\mathbf{b} = \mathbf{1}_d$ : deterministic direction
- SNR  $r = \Theta(1)$  (finite!)

**Idea:** Even with finite SNR, the **large spike in  $\mathbf{W}_1$**  amplifies signals via the quadratic channel.

Related to: one-step gradient descent (Ba et al., 2022), LoRA (Hu et al., 2021).

### Theorem (Spiked Weights)

Define  $\kappa = 1 + r^2/2$ ,  $\kappa_{\perp} = 2 + 2r^2 + r^4/4$ , and  $\beta_y = \frac{3c_{\sigma}^2}{4}\theta_0^4 \varphi \kappa_{\perp}$ .

- (i) **Bulk:**  $ESD \rightarrow \mu$  (same as random weights).
- (ii) **One giant outlier:**  $\hat{\lambda}_1 = b_{\sigma}^2 \theta_0^2 \varphi \kappa \sqrt{N} (1 + o_P(1))$ , eigenvector  $\perp \mathbf{y}$ .
- (iii) **Inherited  $O(1)$  spikes:** All outliers from finite SNR case persist ( $\perp \mathbf{y}$ ).
- (iv) **Additional  $O(1)$  spike:** If  $z'(s_y) > 0$  where  $\beta_y T(s_y) = 1$ , a new outlier emerges.
- (v) **Label alignment:** In the supercritical case,

$$\frac{1}{n} \left| \hat{\mathbf{v}}_y^{\top} \mathbf{y} \right|^2 \xrightarrow{P} \frac{s_y z'(s_y)}{\beta_y T'(s_y)} \cdot \frac{r^4/4}{\kappa_{\perp}} > 0$$

### Key Takeaway

Pretrained weights create a **quadratic feature channel** even at finite SNR. The large spike in  $\mathbf{W}_1$  amplifies the XOR signal through the second-order Hermite component  $c_\sigma$ .

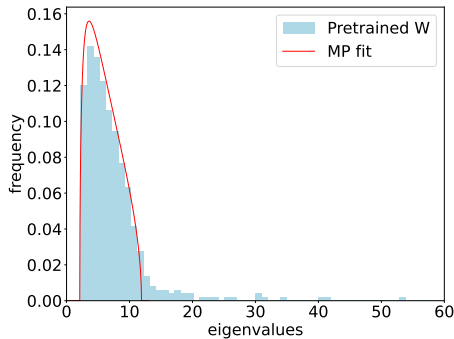
### Interpretation:

- The LE is no longer sufficient: a **Quadratic Equivalent** is needed
- The spike in  $\mathbf{W}_1$  “boosts” the quadratic channel to  $O(1)$  strength
- If  $c_\sigma = 0$ , no label alignment occurs (quadratic channel absent)

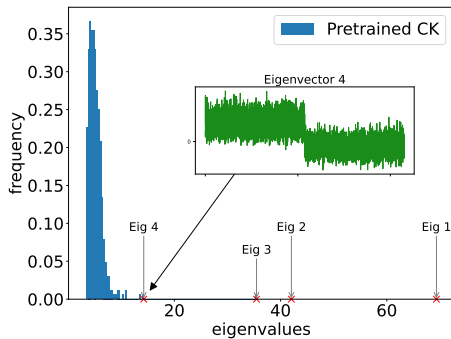
**Empirical verification:** Using weights from a 20-layer NN pretrained on CIFAR-2:

- After sufficient training, 4th eigenvector of CK aligns with XOR labels

## Knob: Pretrained Weights — CIFAR-2 Experiments



ESD of FC1 weights (40 epochs). MP overlaid in red.

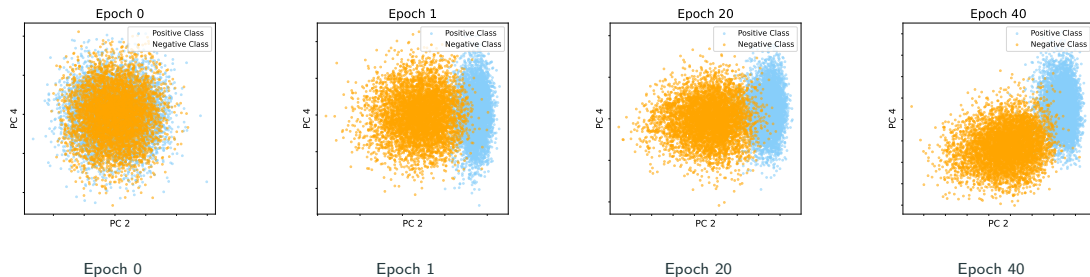


ESD of CK with pretrained weights.

**20-layer NN** trained on CIFAR-2 (88% accuracy, 40 epochs). CK tested on XOR data.  
 $n = 10000$ ,  $d = 3072$ ,  $N = 512$ ,  $r = 5.25$ ,  $\sigma \propto \text{ReLU}$ .

**Key:** Pretrained CK exhibits **4 spikes**, with 4th eigenvector aligning with XOR labels.

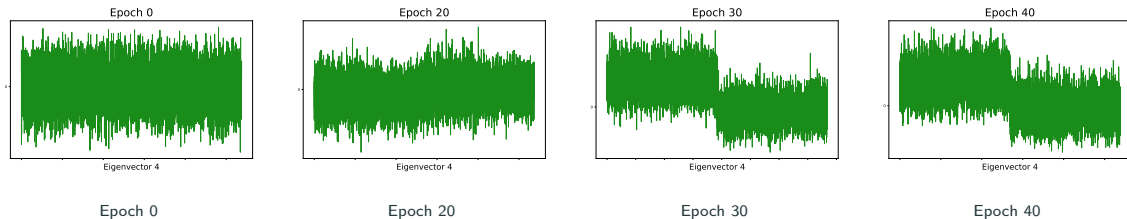
# Knob: Pretrained Weights — Phase Transition Over Training



**PC 2 vs PC 4** of CK with trained weights, tracked from Epoch 0 to 40. Yellow:  $-1$ , Blue:  $+1$ .

**Phase transition:** 4th eigenvector transitions from **pure noise**  $\rightarrow$  **near-perfect XOR alignment**.

# Knob: Pretrained Weights — Eigenvector Alignment



**4th leading eigenvector** of CK with pre-trained weights over training.

**Sharp phase transition** between Epoch 20 and 30: eigenvector aligns perfectly with XOR labels  $y$ .

⇒ **Spectral transition** in the CK parallels the feature learning dynamics in the NN.

## Case 4: Quadratic Sample Size

**Motivation:** In proportional regime  $n = \Theta(d)$ , the LE washes out higher-order structure. What if we have **more data**?

**Regime:**

$$\frac{2n}{d^2} \rightarrow \gamma \in (0, \infty), \quad \frac{n^2}{N} \rightarrow 0 \quad (\text{ultra-wide})$$

**Key assumptions:**  $b_\sigma = 0$  (no linear channel),  $c_\sigma \neq 0$ , SNR  $r = O(1)$ .

**Intuition:** With  $n \asymp d^2$  samples, the CK concentrates around a **quadratic polynomial kernel**:

$$\mathbf{K}^{\text{CK}} \approx a_\sigma^2 P + b_\sigma^2 \mathbf{X}^\top \mathbf{X} + \frac{c_\sigma^2}{2} (\mathbf{X}^\top \mathbf{X}) \odot (\mathbf{X}^\top \mathbf{X})$$

The Hadamard square  $(\mathbf{X}^\top \mathbf{X})^{\odot 2}$  provides the quadratic features needed for XOR.

### Theorem (Quadratic Sample Size)

Let  $\ell = r^4/4$ ,  $\mathbf{K}_c^{\text{CK}} = P\mathbf{K}^{\text{CK}}P$  (centered CK), and  $\lambda_+(\gamma) = (1 + \sqrt{\gamma})^2$ .

- (i) **Bulk:** ESD of  $\mathbf{K}_c^{\text{CK}} \rightarrow \rho_\gamma^{\text{MP}}$ .
- (ii) **BBP for informative outlier:** If  $\ell > 1/\sqrt{\gamma}$ , then

$$\hat{\lambda}_1(\mathbf{K}_c^{\text{CK}}) \xrightarrow{P} (1 - c_\sigma^2/2) + c_\sigma^2 \left( 1 + \gamma + \gamma\ell + \frac{1}{\ell} \right).$$

If  $\ell \leq 1/\sqrt{\gamma}$ , then  $\hat{\lambda}_1(\mathbf{K}_c^{\text{CK}}) \xrightarrow{P} \lambda_+(\gamma)$  (no outlier).

- (iii) **Eigenvector alignment with label:** If  $\ell > 1/\sqrt{\gamma}$ ,

$$\frac{1}{n} \left| \langle \hat{\mathbf{v}}_1(\mathbf{K}_c^{\text{CK}}), \mathbf{y} \rangle \right|^2 \xrightarrow{P} \frac{\gamma\ell^2 - 1}{\gamma\ell(\ell + 1)} \in (0, 1)$$

### Key Takeaway

With quadratic sample complexity  $n \asymp d^2$ , the CK reduces to a quadratic polynomial kernel. Under the BBP threshold  $\ell > 1/\sqrt{\gamma}$ , an informative outlier emerges with **positive label alignment**.

**Phase transition:** The threshold  $\ell > 1/\sqrt{\gamma}$ , i.e.,  $r^4/4 > 1/\sqrt{\gamma}$ , is a sharp condition on the SNR for the kernel to “see” the XOR structure.

### Additional assumptions:

- Ultra-wide network  $N \gg n^2$ : random kernel concentrates around a deterministic kernel
- $b_\sigma = 0$ : removes rank- $d$  linear spikes from  $\mathbf{X}^\top \mathbf{X}$
- Extending to  $N \asymp n$  (linear width) is future work